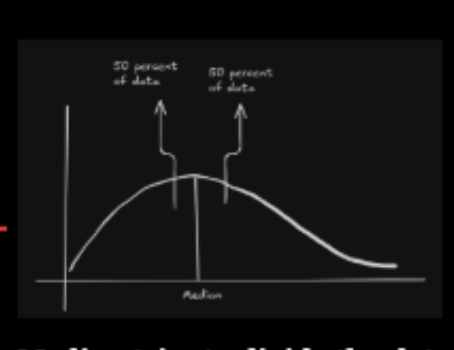
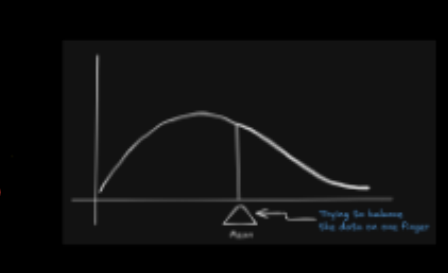
Statistics

- Notes pdf
- Maths Roadmap for Data science
- Resources
- cheat sheets
  - 1
  - 2
  - 3
  - 4
  - 5
- Types
  - Descriptive — description of the data - out of data
  - Inferential — Prediction about the population - out of data
- Basics
  - Population Vs Sample
    - Population
      - The whole group you care about.
      - Example: All people in India.
    - Sample
      - A small part of that group you actually collect data from.
      - Example: 1,000 people surveyed in India.
      - Rules to get sample out of population
        - Sample size should be big enough
        - Sample should be random
        - Sample should represent everyone
    - You study the sample because you can't study the entire population.
- Descriptive
  - Measure of central tendency
    - Mean and Median
      - 50 Percentile == median
      - Median
        - Sort the values and take the middle one (if even 2 people at middle take the avg of it)



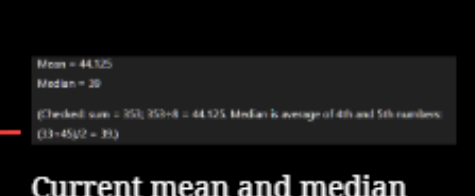
        - Median tries to divide the data so that bothe end has exect 50 percent
      - Mean
        - Sum of all and divide thenm by count
        - Also called as first moment and center gravity of data



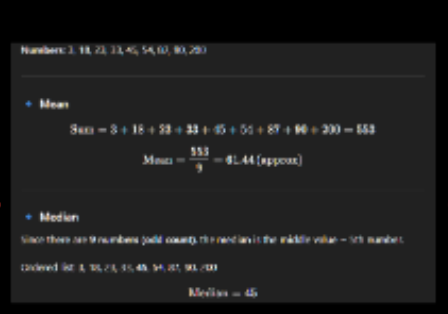
        - Think of a mean as a point on which i am trying to balance the data
        - Mean will push more towards where big values are

Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$
N = number of items in the population	n = number of items in the sample

        - Formula
      - Mean is effected by outliers very largely but the median very less
        - Let suppose we have people from different age -> 3, 18, 23, 33, 45, 54, 87, 90


Mean = 44.125
Median = 39
(Checked: sum = 353; 353÷8 = 44.125. Median is average of 4th and 5th numbers: (33+45)/2 = 39.)

          - Current mean and median
        - Now let suppose suddenly 1 outlier came with the age of 200 year


Numbers: 3, 18, 23, 33, 45, 54, 87, 90, 200

- Mean

$$Sum = 3 + 18 + 23 + 33 + 45 + 54 + 87 + 90 + 200 = 553$$

$$Mean = \frac{553}{9} = 61.44 \text{ (approx)}$$

- Median

Since there are 9 numbers (odd count), the median is the middle value → 5th number.

Ordered list: 3, 18, 23, 33, 45, 54, 87, 90, 200

Median = 45

          - Now mean and median is — Mean effected greatly but not the median
        - Mean will be - average age of a person
        - Median will be - age of a average person
      - IF
        - mean > median — Distribution has more values in the right== right skewed data
        - mean < median — Distribution has more values in the left == left skewed data
        - For semantic distribution mean and median will be the same
    - Mode
      - Most frequent value in dataset
      - used in categorical columns
  - Measure of Dispersion
    - It tells you how spread out the data is
      - If numbers are close together → low dispersion.
      - If numbers are far apart → high dispersion.
    - Types
      - Range
        - Formula → Max – Min
        - Example: In a test, highest = 95, lowest = 45 → Range = 50.
        - Highly effected by outliers, Thats why it is not a good measure to find out the spread of data
      - Variance

$$\sigma^2 = \frac{\Sigma (x-\mu)^2}{N} \quad \text{Population Variance}$$

$$s^2 = \frac{\Sigma (x-\bar{x})^2}{n-1} \quad \text{Sample Variance}$$

        - Average of squared differences from mean.
        - **Used to Tell the spread of the data, but it is squared and not in same unit SD is uded**
        - Measure average distance of each data point form the mean
        - Example: If scores are spread out, variance is big; if close to mean, variance is small.
        - Highly effected by outliers. As we doing square in the formula
      - Mean Absolute Deviation (MAD)

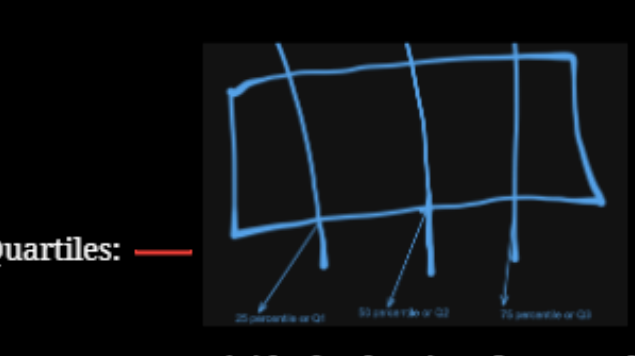
$$MAD = \frac{\sum |x_i - \overline{x}|}{n}$$

        - Same like variance but it does not use square, it use mod.
        - Less effected by outliers like we are doing square here.
        - we can not do prediction over population using MAD of the sample. -> thats why not used very much
      - Standard Deviation (SD)
        - Formula → √Variance
        - **Used to Tell the spread of the data, in the same unit as data**
        - Example: If SD = 0 → all values are the same.
        - Most popular — easy to interpret since it's in the same unit as data.
      - Coefficient of variation (CV)

Formula:

$$CV = \frac{\text{Standard Deviation (SD)}}{\text{Mean}} \times 100$$

        - Used to compare the
        - Problem
          - i have 2 columns age and experience. Now i can not compare the spread of both the columns.
          - Solution — We can use CV to find out which column has more value closer to mean
        - Interpretation
          - Smaller CV → values are closer to the mean → data is more consistent.
          - Larger CV → values more scattered relative to mean → less consistent.
      - Quantiles and Percentiles / 5 Number summary
        - Quantiles
          - are statistical measures used to divide a set of numerical data into equal-sized groups, with each group containing an equal number of observations.
          - Types
            - Quartiles:



              - Divide the data into four equal parts, Q1 (25th percentile), Q2 (50th percentile or median), and Q3 (75th percentile).
            - Deciles: — Divide the data into ten equal parts, D1 (10th percentile), D2 (20th percentile), ..., D9 (90th percentile).
            - Percentiles: — Divide the data into 100 equal parts, P1 (1st percentile), P2 (2nd percentile), ..., P99 (99th percentile).
            - Quintiles: — Divides the data into 5 equal parts
        - **Things to remember while calculating these measures:**
          - 1. Data should be sorted from low to high
          - 2. You are basically finding the location of an observation
          - 3. They are not actual values in the data
          - 4. All other tiles can be easily derived from Percentiles
        - **Percentiles**
          - A percentile is a statistical measure that represents the percentage of observations in a dataset that fall below a particular value. — A percentile tells you the value below which a certain % of data falls.
          - Suppose your exam score is at the 90th percentile. — → This means you scored better than 90% of students.
          - Formula

$$PL = \frac{P}{100}(N+1)$$

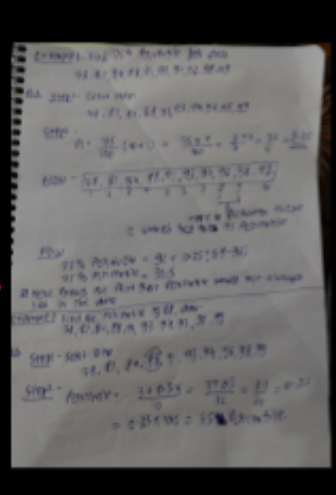
            - Formula to calculate the percentile value:
              - PL = the desired percentile value location
              - N = the total number of observations in the dataset
              - p = the percentile rank (expressed as a percentage)

Percentile of a value

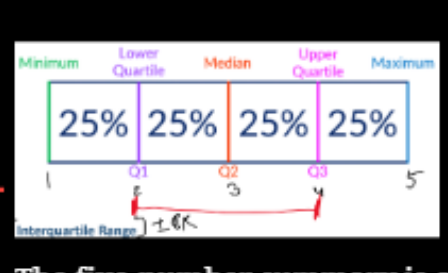
$$\text{Percentile rank} = \frac{X + 0.5Y}{N}$$

X = number of values below the given value
Y = number of values equal to the given value
N = total number of values in the dataset

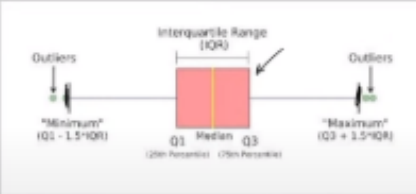
            - Percentile of a value
              - X = number of values below the given value
              - Y = number of values equal to the given value
              - N = total number of values in the dataset



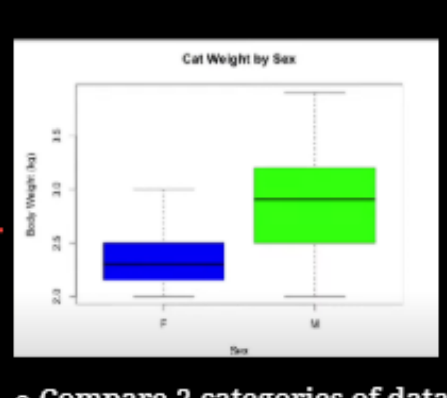
          - Exmaples
        - **5 number summary**
          - The five-number summary includes the following values:
            - Minimum value
            - First quartile (Q1) - 25 Percentile
            - Median (Q2) - 50 Percentile — Between 25 to 75 percentile is called IQR (Interquartile range)
            - Third quartile (Q3) - 75 Percentile
            - Maximum value



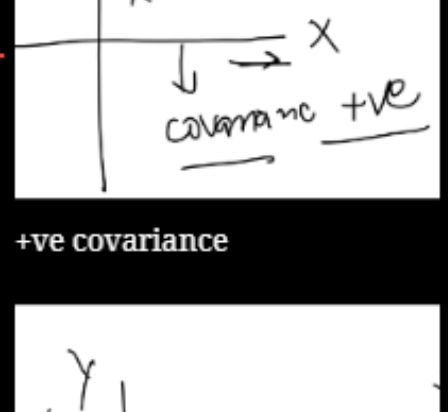
          - The five-number summary is often represented visually using a box plot



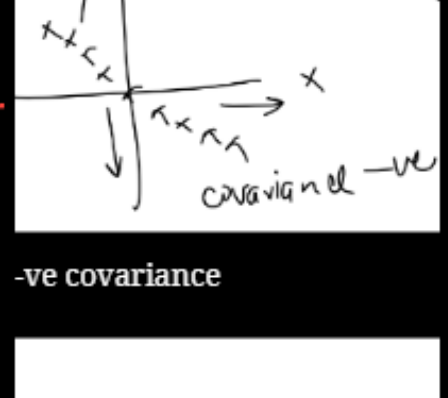
            - BOX PLOT
          - Benefits of a Boxplot
            - o Easy way to see the distribution of data
            - o Tells about skewness of data
            - o Can identify outliers



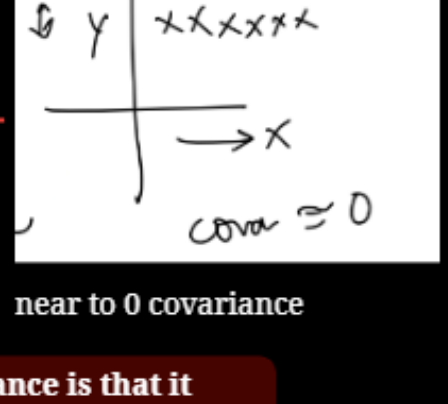
            - o Compare 2 categories of data
  - Graphs on the basis of data types
    - univariate analysis
      - categorical columns
        - frequency distribution table
        - pie chart
        - bar chart
      - numerical columns
        - histogram
        - KDE
    - univariate analysis
      - categorical - categorical — Crosstab/Contingency table
      - numerical - categorical
        - bar chart
        - crosstab (if we divide the numerical column in bins)
      - numerical - numerical — Scatter plot
  - Covariance
    - **It tells us how two variables change together.**
    - **It's the foundation of correlation (which is just standardized covariance).**
    - **If covariance value is positive = positive relationship,**
If covariance value is negative = negative relationship
If covariance value is zero = no relationship
      - Example 1->



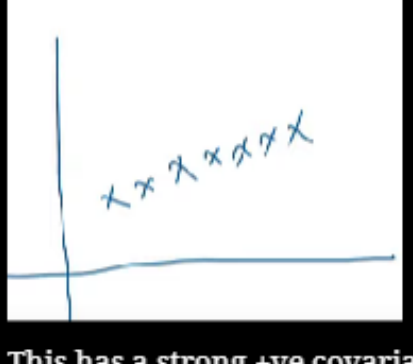
        - +ve covariance
        - Reason -> on increasing value of column X value of column Y also increases
      - Example 2->



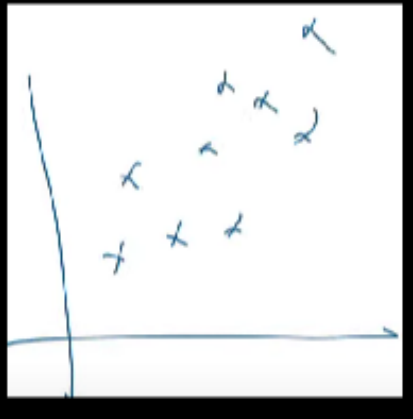
        - -ve covariance
        - Reason -> on increasing value of column X value of column Y decreases
      - Example 3->



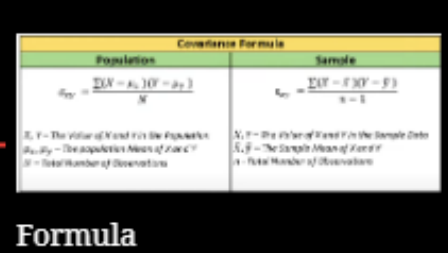
        - near to 0 covariance
        - Reason -> on increasing value of column X value of column Y does not change or
        - if on increasing value of column X value of column Y does not predictable.
    - Disadvantages of using Covariance
      - **One limitation of covariance is that it does not tell us about the strength of the relationship between two variables, since the magnitude of covariance is affected by the scale of the variables.**
      - **This problem solved by correlation**
      - Example



        - This has a strong +ve covariance



        - This has a weak +ve covariance
        - But covariance not able to tell about the strength of relation, it only tell us about +,- or 0



Covariance Formula
Population
Sample

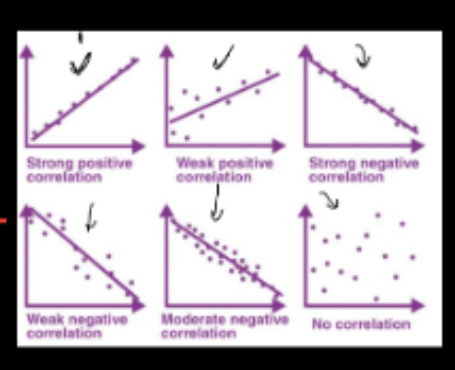
    - Formula — **Important interview question** — **Covariance of a variable with itself ?**

$$\sum_{i=1}^{n} \frac{(x-\bar{x})(y-\bar{y})}{n-1} = \sum_{i=1}^{n} \frac{(x-\bar{x})(x-\bar{x})}{n-1} = \boxed{\sum_{i=1}^{n} \frac{(x-\bar{x})^2}{n-1}}$$

      - It will become variance
  - **Correlation**
    - Correlation tells us how strong and in what direction two variables are related.
    - It is like covariance but with magnitude
    - Does not depend on scale of data like covariance
    - "correlation does not imply causation"

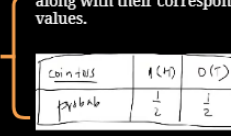
$$Correlation = \frac{Cov(x,y)}{\sigma x * \sigma y}$$

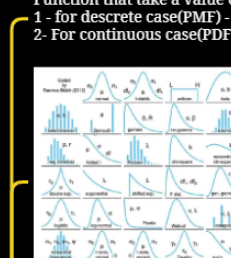
    - Formula

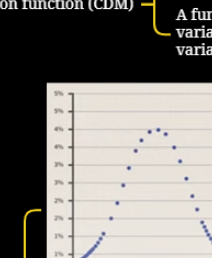
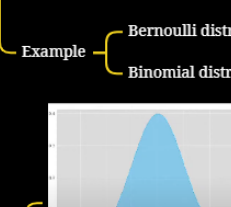


    - Its value always lie in between -1 t0 1
      - if value is negative = that mean negative correlation
      - if value is positive = that mean positive correlation
      - if value is 0 = that mean no correlation
      - more the value closer to 1 -> mean stronger + correlation
      - more the value closer to -1 -> mean stronger - correlation

Statistics

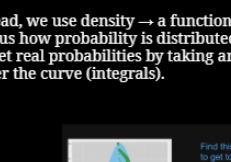
- **Random Variables**
  - A Random Variable is a set of possible values from a random experiment.
  - Types
    - Discrete Random Variable
    - Continuous Random Variable
  - what is a random experiment
    - Random experiment with Discrete random variable
      - rolling a dice
      - tossing a coin
    - Random experiment with Continuous random variable — Asking a marks of a student
  - Example
    - Random variable for tossing a coin — { 0,1 }
    - Random variable Asking a marks of a student — {100 ..... 71.9 ..... 0}
- **Probability Distributions**
  - A probability distribution is a list of all of the possible outcomes of a random variable along with their corresponding probability values.
  - 

    Example - probability Distribution for a coin toss
- **Probability Distributions Function**
  - Function that take a value of a random variable and return the
    1 - for descrete case(PMF) - probability of that value
    2- For continuous case(PDF) - probabilty density at the value
  - 

    Famous Probability Distributions
  - Type
    - Probability Mass function (PMF) — A function that take value of a discrete random variable and return probability of that value
    - Probability density function (PDF) — A function that take value of a continuous random variable and return probability density at the value
    - Cumulative distribution function (CDM)
      - created from PMF or PDF
      - A function that takes a value of a random variable and returns the probability that the variable is less than or equal to that value.
  - Probability Mass Function (PMF)
    - 
    - A function that take value of a discrete random variable and return probability of that value
    - Here x axis = value of column and y axis = probability
    - The probabilities assigned by the PMF must satisfy two conditions:
      - The probability assigned to each value must be non-negative (i.e., greater than or equal to zero).
      - The sum of the probabilities assigned to all possible values must equal 1.
    - Example
      - Bernoulli distribution
      - Binomial distribution
  - **Probability density function (PDF)**
    - 
    - A function that take value of a continuous random variable and return probability density at the value
    - **Here x axis = value of column and y axis = probability density**
    - **Q1 -Why Probability Density and why not Probability? on y - axis**
      - Unlike the discrete random variable which has countable number of values

        In the continuous variable have - infinite number of values.

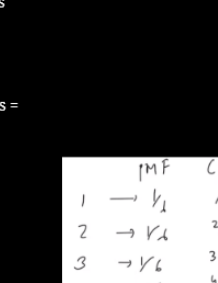
        The probability of hitting exactly one point (e.g., height = 170.000000...) is 0.

        Instead, we use density → a function that tells us how probability is distributed, and we get real probabilities by taking areas under the curve (integrals).
    - **Q2 -What does the area of this graph represents?**
      - Probability!! — 

        Example- probability of getting marks in between 5-6 = will be tell by calculating the area under probability density(5-6). that will be the probability
      - we can not get the probability at a fix point like we can not calculate the probability at 6, but we can calculate the area between 6-6.01 or 6-6.0001 , and the area will represent the probility of happeining.
    - **Note -> if you integrate Probability density function (PDF) it will give you Cumulative Distribution Function(CDF)**

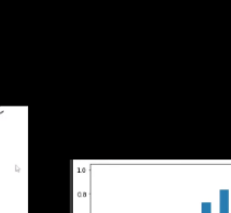
      and if you differentiate Cumulative Distribution Function(CDF) it will give you Probability density function (PDF)
    - The Total area under the curve = 1
  - Cumulative Distribution Function(CDF)
    - The cumulative distribution function (CDF) F(x) describes the probability that a random variable X with a given probability distribution will be found at a value less than or equal to x

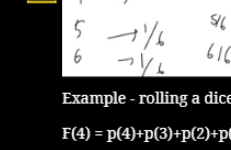
      F(x) = P(X <= x)

      Here x axis = value of column and y axis = probability
    - Cumulative Distribution Function(CDF) of PMF
      - 

	PMF	CDF
1	→ 1/6	1/6
2	→ 1/6	2/6
3	→ 1/6	3/6
4	→ 1/6	4/6
5	→ 1/6	5/6
6	→ 1/6	6/6

        Example - rolling a dice ->

        F(4) = p(4)+p(3)+p(2)+p(1)
      - 

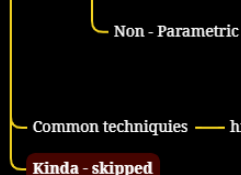
        cdf graph
    - Cumulative Distribution Function(CDF) of PDF
      - 

        Example - rolling a dice ->

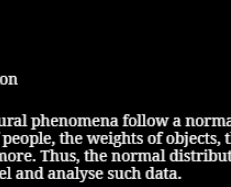
        F(4) = p(4)+p(3)+p(2)+p(1)
  - Density Estimation
    - why we are learning Density Estimation — Use to estimate the probability distribution (PDF) from a random variable based on data
    - types
      - Parametric DE
        - Here we assume the data follows a specific distribution shape (like Normal, Exponential, etc.).

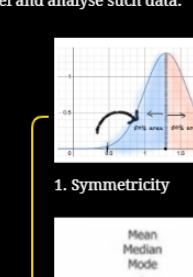
          Then we only need to estimate the parameters (like mean μ and variance σ² for Normal distribution).

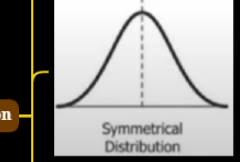
          Once parameters are found, the whole distribution (PDF) is defined.
        - Example — Gaussian mixture models (GMMs)
      - Non - Parametric DE
        - Here we don't assume the data follows a specific distribution shape (like Normal or Exponential).
        - Instead, we let the data itself decide the shape.

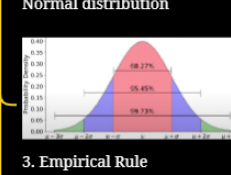
          It's more flexible but can be more complex.
        - Example — **Kernel density estimation (KDE)** — **skipped**
    - Common techiniques — histogram estimation
    - **Kinda - skipped**
  - **Normal Distribution or Bell curve**
    - 

      It is a continuous probability distribution (it is a pdf) that is symmetrical around the mean, with a bell-shaped curve.
    - $$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

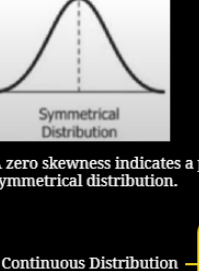
      PDF Equation of Normal Distribution
    - Commonality in Nature: Many natural phenomena follow a normal distribution, such as the heights of people, the weights of objects, the IQ scores of a population, and many more. Thus, the normal distribution provides a convenient way to model and analyse such data.
    - **Properties of Normal Distribution**
      - 

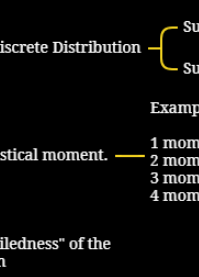
        1. Symmetricity
      - 

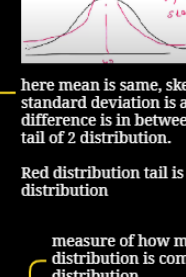
        mean == median == mode , For a proper Normal distribution
      - 

        3. Empirical Rule
        - The normal distribution has a well-known empirical rule, also called the 68-95-99.7 rule, which states that approximately 68% of the data falls within one standard deviation of the mean, about 95% of the data falls within two standard deviations of the mean, and about 99.7% of the data falls within three standard deviations of the mean.
      - 4. The area under the curve = 1
    - **Standard Normal Variate or Standard Normal Distribution (Z)**
      - 

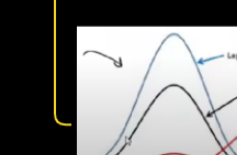
        A Standard Normal Variate(Z) is a standardized form of the normal distribution with mean = 0 and standard deviation = 1.
      - $$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

        Equation
  - **Skewness**
    - Skewness is a measure of the asymmetry of a probability distribution.
    - **In a symmetrical distribution, the mean, median, and mode are all equal. In contrast, in a skewed distribution, the mean, median, and mode are not equal, and the distribution tends to have a longer tail on one side than the other.** — **The greater the skew the greater the distance between mode, median and mean.**
    - Types
      - -ve / Left
        - 

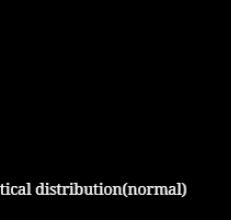
          a negative skewness means that the tail is longer on the left side.
        - mean<median<mode
      - +ve / Right
        - 

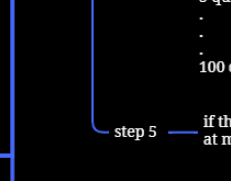
          A positive skewness means that the tail of the distribution is longer on the right side.
        - mode<median<mean
      - 0
        - 

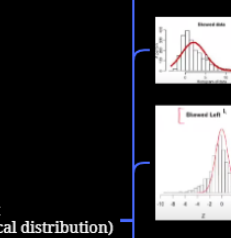
          A zero skewness indicates a perfectly symmetrical distribution.
  - Some other Distribution
    - other Continuous Distribution
      - uniform Distribution
      - log normal Distribution
      - Pareto Distribution
    - other Discrete Distribution
      - Subtopic 1
      - Subtopic 2
  - Kurtosis
    - Kurtosis is the 4th statistical moment. — Example -

      1 moment = mean
      2 moment = std
      3 moment = skewness
      4 moment = Kurtosis
    - is a measure of the "tailedness" of the probability distribution
    - Example -> — 

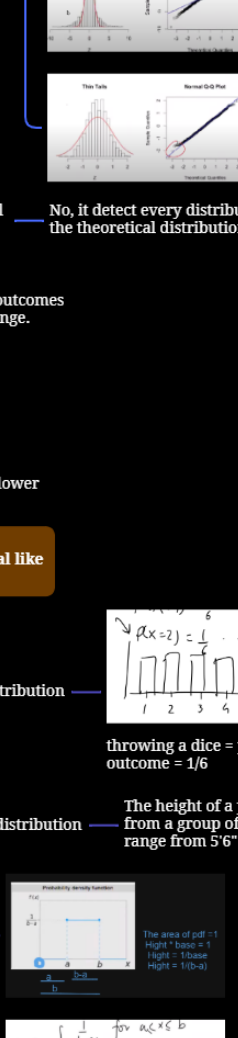
      here mean is same, skewness is same , standard deviation is also the same but the difference is in between the fatness of the tail of 2 distribution.

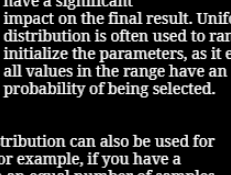
      Red distribution tail is mor fater then black distribution
    - Excess Kurtosis
      - measure of how much more peaked or flat a distribution is compared to a normal distribution,
      - How it is calculate excess kurtosis
        - step1 - — considered normal distribution kurtosis as 0.
        - step 2 - — calculate sample kurtosis of the distribution
        - step 3 - — sample kurtosis -3
          - < 0 — Leptokurtic
          - > 0 — Platykurtic
          - = 0 — Mesokurtic
      - 

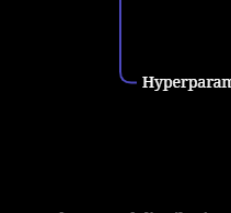
        Types
        - Leptokurtic
          - A distribution with positive excess kurtosis is called leptokurtic.
          - In terms of shape, a leptokurtic distribution has fatter tails.
          - This indicates that there are more extreme values or outliers in the distribution.
        - Platykurtic
          - A distribution with negative excess kurtosis is called platykurtic.
          - a platykurtic distribution has thinner tails.
          - This indicates that there are fewer extreme values or outliers in the distribution.
        - Mesokurtic
          - Distributions with zero excess kurtosis are called mesokurtic.
          - indicating that it has the same degree of "peakedness" or "flatness" as a normal distribution.
- **How to find if a given distribution is normal or not?**
  - WAYS
    - Visual inspection: — histogram or a density plot of the data.
    - QQ Plot:
    - Statistical tests:
      - Shapiro-Wilk test,
      - Kolmogorov-Smirnov test.
      - Anderson-Darling test,
- QQ Plot
  (quantile-quantile plot)
  - A QQ plot (quantile-quantile plot) is a graphical tool used to assess the similarity of the distribution of two sets of data.
  - 

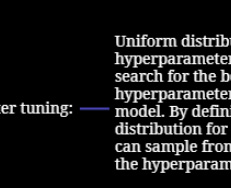
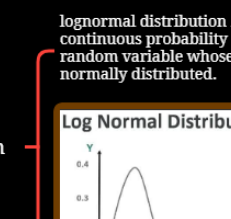
    Example
  - Steps
    - step 1 — Get a theoretical distribution(normal)
    - step 2
      - sort x(your data)
      - sort y(theoretical distribution)
    - step 3 — calculate quantiles of x and y
    - step 4 — 

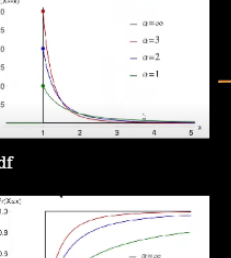
      start plotting quantiles of x and y over the graph like ->

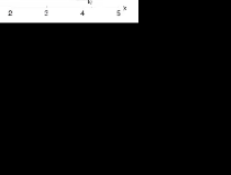
      1 quantile of x with 1 quantile of y
      2 quantile of x with 2 quantile of y
      3 quantile of x with 3 quantile of y
      .
      .
      100 quantile of x with 100 quantile of y
    - step 5 — if there is a straight line over all the points at mean they are similar
  - Interpret QQ plot
    (normal theoretical distribution)
    - 
  - Does QQ plot only detect normal distribution? — No, it detect every distribution based upon the theoretical distribution you have choose
- Uniform distribution
  - probability distribution where all outcomes are equally likely within a given range.
  - Types
    - Discrete
    - Continuous
  - $X \sim U(a,b)$

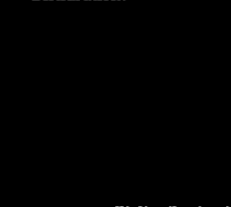
    notation -> where a,b = upper and lower limit and X= data set
  - **skewness of continuous uniform distribution = 0 = it is symmetrical like normal distribution**
  - Example
    - discrete uniform distribution — 

      throwing a dice = probability of each outcome = 1/6
    - continuous normal distribution — The height of a person randomly selected from a group of individuals whose heights range from 5'6" to 6'0"
  - 

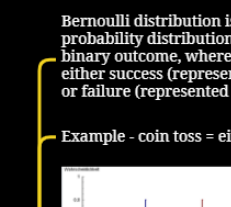
    pdf
    - 
    - $$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \le x \le b \\ 0 & \text{for otherwise} \end{cases}$$
  - 

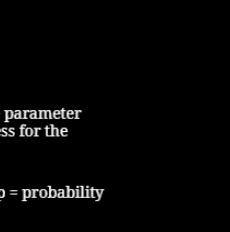
    cdf
  - **Example**
    - Random initialization: — In many machine learning algorithms, such as neural networks and k-means clustering, the initial values of the parameters can have a significant impact on the final result. Uniform distribution is often used to randomly initialize the parameters, as it ensures that all values in the range have an equal probability of being selected.
    - Sampling: — Uniform distribution can also be used for sampling. For example, if you have a dataset with an equal number of samples from each class, you can use uniform distribution to randomly select a subset of the data that is representative of all the classes.
    - Hyperparameter tuning: — Uniform distribution can also be used in hyperparameter tuning, where you need to search for the best combination of hyperparameters for a machine learning model. By defining a uniform prior distribution for each hyperparameter, you can sample from the distribution to explore the hyperparameter space.
- Log normal distribution
  - lognormal distribution is a heavy tailed continuous probability distribution of a random variable whose logarithm is normally distributed.
  - 

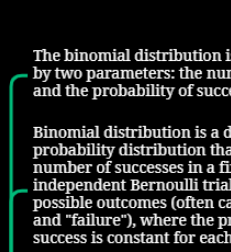
    a right skewed distribution is called log normal distribution , if we take log of the data makes a normal distribution
- Pareto distribution
  - Power law
    - 

      a power law is a functional relationship between two variables, where one variable is proportional to a power of the other.
    - y = k * x^a — where k= some random constant y,x= variable and a = power
  - The Pareto distribution is a type of probability distribution that is commonly used to model the distribution of wealth, income, and other quantities that exhibit a similar power-law behaviour
  - 

    pdf — $$\frac{\alpha x_m^\alpha}{x^{\alpha+1}}$$

    Pdf equation
  - 

    cdf
  - How to detect if a distribution is Pareto Distribution?
    - by the help of log log plot
      - y = $$\frac{\alpha x_m^\alpha}{x^{\alpha+1}}$$
      - 

        take log of both x and y and then plot a graph of it , if looks like this then it is a pareto distribution
    - QQ plot — instant of taking a normal distribution take pareto distribution and then compare both of them
- Bernoulli Distribution
  - Bernoulli distribution is a discrete probability distribution that models a binary outcome, where the outcome can be either success (represented by the value 1) or failure (represented by the value 0).
  - Example - coin toss = either head or tail
  - 

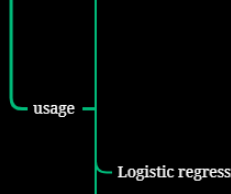
    Pmf graph
  - $$f(k;p) = p^k(1-p)^{1-k}$$

    Pmf equation
    - this distribution has only one parameter which p= probability of success for the case(k)
    - here k= out come you want , p = probability of k
    - Example :
      - $P(X=1) = \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^0$

        coming 1 on toss
      - $P(X=0) = \left(\frac{1}{2}\right)^0 \left(1-\frac{1}{2}\right)^{1-0} = 1 \times \frac{1}{2}$

        coming 0 on toss
- Binomial Distribution
  - The binomial distribution is characterized by two parameters: the number of trials n and the probability of success p.

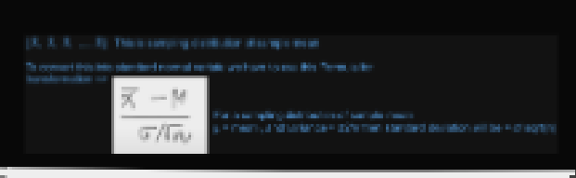
    Binomial distribution is a discrete probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials with two possible outcomes (often called "success" and "failure"), where the probability of success is constant for each trial.
  - Example
    - **deciding a mail is spam is Bernoulli Distribution but it deciding a mail is spam on the basis of 10 mails is Binomial Distribution**
    - **if the number of trials n == 1 and then this Binomial Distribution will become bernoulli distribution**
  - Formula — $P(X) = {}_nC_x\, p^x (1-p)^{n-x}$

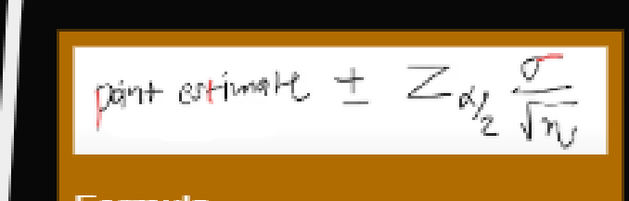
    n = number of tiles
    p = probability of success
    x = desired result
  - The Probability of anyone watching this lecture in the future and then liking it is 0.5. What is the probability that:
    - No-one out of 3 people will like it
    - 1 out of 3 people will like it
    - 2 out of 3 people will like it
      - 3C2*(0.5)^2*(0.5)^(3-2)
      - 3/8
    - 3 out of 3 people will like it
  - 
  - Criteria:
    - 1. The process consists of n trials
    - 2. Only 2 exclusive outcomes are possible, a success and a failure.
    - 3. P(success) = p and P(failure) = 1-p and it is fixed from trial to trial
    - 4. The trials are independent.
  - usage
    - Binary classification problems: — In binary classification problems, we often model the probability of an event happening as a binomial distribution. For example, in a spam detection system, we may model the probability of an email being spam or not spam using a binomial distribution.
    - Hypothesis testing: — In statistical hypothesis testing, we use the binomial distribution to calculate the probability of observing a certain number of successes in a given number of trials, assuming a null hypothesis is true. This can be used to make decisions about whether a certain hypothesis is supported by the data or not.
    - Logistic regression: — Logistic regression is a popular machine learning algorithm used for classification problems. It models the probability of an event happening as a logistic function of the input variables. Since the logistic function can be viewed as a transformation of a linear combination of inputs, the output of logistic regression can be thought of as a binomial distribution.
    - A/B testing: — A/B testing is a common technique used to compare two different versions of a product, web page, or marketing campaign. In A/B testing, we randomly assign individuals to one of two groups and compare the outcomes of interest between the groups. Since the outcomes are often binary (e.g., click- through rate or conversion rate), the binomial distribution can be used to model the distribution of outcomes and test for differences between the groups.

Statistics3

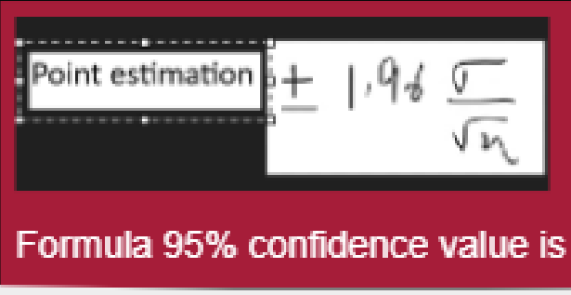
- **Sampling Distribution**
  - For example ->

    salary of a Indian
    - First we decide the size of sample we gone choose from Indian population = 50
    - then we get a sample of 50 people as from population
      - S1
        - 0, 10k , 50k , ,,,,,,,,,, 5k
    - similarly we gone take 100 samples
      - S2
      - S3
      - .
      - .
      - .
      - S100
    - Now we gone get the mean of each sample == which will know as **Sample mean**
      - S1
        - mean = M1
      - S2
        - mean = M2
      - S3
        - mean = M3
      - .
        - .
      - .
        - .
      - .
        - .
      - S100
        - mean = M100
    - Now all these mean combined know as = sampling distribution of sample mean
      - [M1 , M2 , M3 .... M100]
    - If we have taken variance instant of variance then it will known as = sampling distribution of sample variance
  - Sampling distribution is a probability distribution that describes the statistical properties of a sample statistic (such as the sample mean or variance) computed from multiple independent samples(100 in above example) of the same size(50 in above example) from a population.
- **Central Limit Theorem**
  - The Central Limit Theorem (CLT) states that the distribution of the sample means of a large number of independent and identically distributed random variables will approach a normal distribution, regardless of the underlying distribution of the variables.
  - in other words , distribution of sample mean should always follow the normal distribution, regardless of the underlying distribution of the data.
    - Note here the data mean and sample mean remain the same
    - Note here the data variance = sample (variance/n) where n= sample size
  - Now the normal distribution we get , i want to convert this into standard normal variate =

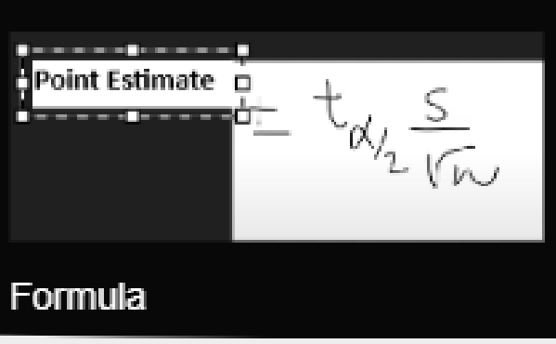
    $$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$
  - conditions
    - 1. The sample size is large enough, typically greater than or equal to 30.
    - 2. The sample is drawn from a finite population or an infinite population with a finite variance.
    - 3. The random variables in the sample are independent and identically distributed.
- **Confidence interval**
  - is a range of value within which we except a particular population parameter like mean to fall
  - confidence level
    - usually expressed as a percentage like 95% indicates how sure we are that the true value lies within interval
    - It is not probability
    - what does confidence level mean -> For example if i have salary in between [10K , 50K] with confidence level 0f 95%
      - if we get the sampling distribution of sample (mean/variance ..) then out of 100 times 95 times sample mean/variance.. will gone be in the range
  - Confidence interval is used for parameters not for statistics. Statistics help us get the confidence interval for a parameter
  - Formula
    - confidence interval = point estimate ± Margin of error
    - EXAMPLE
      - confidence interval = 25 ± 4 = [21 , 29]
  - Ways to calculate Confidence interval
    - Z procedure
      - used when we have std of the population
    - t procedure
      - used when we do not have std of the population
  - **Z procedure / Sigma known**
    - Assumptions
      - Random sampling
      - Known population standard deviation
      - Population should be Normally distributed or sample size larger then 30(so that we can apply central limit theorem)
    - 

      $$\text{point estimate} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

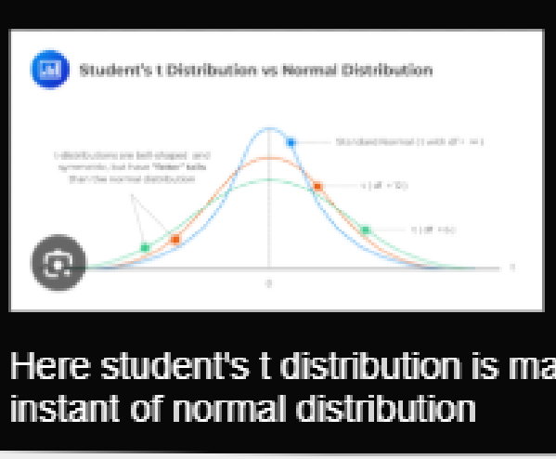
      Formula
      - here Point estimate = can be mean , etc
      - n = sample size
      - σ = std of population
      - Z = critical value
        - Z = z table value of z(α/2)
      - 1- α = confidence level (mostly 95%)
    - 

      $$\text{Point estimation} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

      Formula 95% confidence value is
    - confidence interval is directly propertion to confidence interval
      - Example 100% confident about dhoni gone make run in between [0 -300]
        - 100% confident but highest the range of interval
      - Example 1% confident about dhoni gone make run in between [123 - 125]
        - 1% confident but lower the range of interval
    - Factors Affecting Margin of Error

      margin of error = (upper limit - lower limit)/2
      - 1. Confidence Level
        - directly proportional
        - reason -> more the confidence level , more the interval size , more the margin of error
      - 2. Sample Size
        - inversely proportional
      - 3. Population Standard Deviation
        - directly proportional
  - **t procedure / Sigma not known**
    - Assumptions
      - Random sampling
      - Each observation should be independent
      - Population should be Normally distributed or sample size larger then 30(so that we can apply central limit theorem)
      - Known Sample standard deviation
    - 

      $$\text{Point Estimate} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

      Formula
      - here Point estimate = can be mean , etc
      - n = sample size
      - s = std of sample -> if we are using clt -> then we take a mean of all the sample std Example -> you have taken 10 samples [s1 + s2 .... s10]/10
      - t = critical value
        - t = t table value of t(α/2)
      - 1- α = confidence level (mostly 95%)
    - 

      Here student's t distribution is make instant of normal distribution
- **Hypothesis Testing**
  - A statistical hypothesis test is a method used to decide whether the data at hand sufficiently support a particular hypothesis.
    - For example -

      average watch time of my video is 7 min

      if i make some changes in my video format

      and resealed 5 new video based on new format ->

      average watch time of these video is 7 ,9 , 13 , 5 , 11

      Now to tell the new format is better or not-> we use hypothesis testing
  - Hypothesis testing allows us to make probabilistic statements about population parameters.
  - Terms
    - Null hypothesis/status quo (H0):
      - the null hypothesis is a statement that assumes there is no significant effect or relationship between the variables being studied.
      - Example
        - H0 : new format will not gone increase watch time

          H0 : μ = 7 min
    - Alternative hypothesis/research hypothesis (H1 or Ha):
      - The alternative hypothesis, is a statement that contradicts the null hypothesis and claims there is a significant effect or relationship between the variables being studied.
      - Example
        - H1 : new format will gone increase watch time

          H1 : μ > 7 min
  - important points
    - 1
      - Typically the Null hypothesis says nothing new is happening
    - 2
      - We try to gather evidence to reject the null hypothesis
    - 3
      - failing to reject the null hypothesis doesn't necessarily mean that the null hypothesis is true
  - Approach
    - Rejection Region Approach
      - Steps
        - 1. Formulate a Null and Alternate hypothesis
        - 2. Select a significance level(This is the probability of rejecting the null hypothesis when it is actually true, usually set at 0.05 or 0.01)
        - 3. Check assumptions (example distribution)
        - 4. Decide which test is appropriate(Z-test, T-test, Chi-square test, ANOVA)
        - 5. State the relevant test statistic
        - 6. Conduct the test
        - 7. Reject or not reject the Null Hypothesis.
        - 8. Interpret the result
      - Example
        - watch this video-> time stramp 34:48 to 1:00
    - P value Approach

I am gone use this relation to explain each concept of it!!